

KingsHire (kingshire-v3) — Agent Handover Document

Date produced: 2026-06-26

Purpose: Complete project state handover for AI agents or developers picking up this codebase.

1. What This Project Is

KingsHire is a community-first freelance marketplace for members of Believer's Love World / Christ Embassy.

- **Live URL:** <https://kingshire.uk>
- **Staging URL:** Railway staging environment (auto-deploys from `staging` branch)
- **Two primary user roles:**
 - **Client** — Posts jobs, hires workers, pays via escrow
 - **Kinglancer** — The worker/freelancer who applies for and completes jobs
 - A user can hold both roles simultaneously
- **Admin panel:** `/admin` — protected by `ADMIN_PASSCODE` env var + HTTP-only cookie session

2. Tech Stack

Layer	Technology	Version
Framework	Next.js (App Router)	16.2.4
UI	React	19.2.4
Language	TypeScript	~5.x
Styling	Tailwind CSS	v4 (breaking: use <code>bg-linear-to-*</code> NOT <code>bg-gradient-to-*</code>)
Animations	Framer Motion	^12
Database	Supabase (PostgreSQL)	@supabase/ssr 0.10.3
Auth	Supabase Auth (PKCE) + KingsChat SSO + Google OAuth	—
Payments	Stripe Connect (separate charges/transfers)	—
Email	Brevo (formerly Sendinblue)	REST API via <code>BREV0_API_KEY</code>
Hosting	Railway (persistent server, NOT serverless/edge)	—
Icons	lucide-react	^1.8

Key architectural notes

- **NOT serverless.** Railway runs a persistent Node.js server. In-memory Next.js data cache **persists between requests but is cleared on every deploy/restart.**
- `proxy.ts` is **NOT Next.js middleware.** It is a separate Express proxy server for Railway routing.

- `revalidateTag(tag, { expire: 0 })` is valid in this Next.js version — `CacheLifeConfig = { expire?: number }`.
- `unstable_cache` is used for ISR-style caching.
- `getDashboardContext` is wrapped in `React cache()` for per-render deduplication within a single request.
- **Supabase keys:** Uses the new `NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY` format (not the old `NEXT_PUBLIC_SUPABASE_ANON_KEY`).

3. Supabase Database

Project

- **URL:** <https://mdzousozzrnggtblusws.supabase.co>
- **Project ref:** `mdzousozzrnggtblusws`
- **Service role key:** stored as `SUPABASE_SECRET_KEY` env var

Tables

Table	Purpose
profiles	One per auth user. Role, bio, services (JSONB), stripe info, rating, avatar, etc.
jobs	Job postings. Statuses: open, in_progress, completed, cancelled, disputed, approved.
applications	Kinglancer applications to open jobs.
transactions	Escrow records. One per job (<code>transactions_job_id_unique</code> unique constraint). Statuses: held, released, refunded, transferred.
payment_attempts	Tracks Stripe PaymentIntents before finalization. Replaces older "legacy" direct transactions.
notifications	In-app notifications. Read via <code>NotificationBell</code> component polling.
reviews	Double-blind review system (both parties submit; revealed simultaneously after both submit OR after 14-day timeout).
disputes	Disputes raised on in_progress/completed jobs.

Important DB constraints

- `transactions_job_id_unique` — only ONE transaction per job (prevents double-charging). This is intentional.
- RLS enabled on all tables.
- Service role client (`createServiceClient()` from `lib/supabase/service.ts`) bypasses RLS — used in all API routes and webhook handlers.

Migrations

Files in `supabase/migrations/` — numbered `002` through `026` . **These are NOT auto-applied**. Each must be run manually in the Supabase SQL Editor.

Migration 027 (`supabase/migrations/027_review_query_indexes.sql`) has been created but NOT yet run on staging or production. It adds two performance indexes:

```
create index if not exists idx_reviews_reviewer_job
on public.reviews (reviewer_id, job_id);

create index if not exists idx_reviews_reviewee_published_at
on public.reviews (reviewee_id, published_at desc)
where is_published;
```

Action required: Run this SQL manually in Supabase SQL editor for both staging and production.

4. Environment Variables

All must be set in Railway service environment settings.

Variable	Required	Description
NEXT_PUBLIC_SUPABASE_URL	✓	Supabase project URL
NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY	✓	Supabase anon/publishable key (new format)
SUPABASE_SECRET_KEY	✓	Supabase service role key
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY	✓	Stripe publishable key
STRIPE_SECRET_KEY	✓	Stripe secret key
STRIPE_WEBHOOK_SECRET	✓	Stripe webhook signing secret
NEXT_PUBLIC_APP_URL	✓	Full base URL (e.g. https://kingshire.uk)
BREVO_API_KEY	✓ for email	Brevo (email provider) API key
BREVO_SENDER_EMAIL	✓ for email	From address (e.g. noreply@kingshire.uk)
BREVO_SENDER_NAME	optional	Display name (e.g. KingsHire)
ENABLE_EMAIL	✓ for email	Must be true to send any emails
ADMIN_PASSCODE	✓	Password for admin panel login
ADMIN_SESSION_SECRET	✓	Secret for signing admin HTTP-only cookie
ADMIN_NOTIFICATION_EMAIL	optional	Admin email for dispute notifications
CRON_SECRET	✓	Bearer token for cron job endpoints
KINGSCHAT_CLIENT_ID	✓	KingsChat OAuth client ID
KINGSCHAT_API_KEY	✓	KingsChat API key
APP_ENV	optional	staging or production

5. Branch / Deployment State

Branch	Railway Service	Current State
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main	Production (kingshire.uk)	At commit 7756485 — does NOT include email fixes or webhook idempotency fix
staging	Staging	At commit 458019a — includes email fixes + webhook idempotency fix

Staging-only commits (not yet on `main`)

In order (oldest first):

1. 97c332b — "Surface pending reviews as Action Centre items with countdown"
2. 523087a — **fix: send email notifications when a job is posted**
3. ef5fbee — **fix: client application notification links directly to the job**
4. 458019a — **fix: idempotent webhook handling for duplicate payment_intent.succeeded** ← URGENT

All 4 commits need to be merged to `main` for production. Commit `458019a` is the most urgent (active production bug).

Feature work NOT on any branch

The following features were built in a previous session, then the staging branch was hard-reset to match production. These changes exist only in the previous session's git history and need to be **re-applied**:

- Profile completeness gating (onboarding bio requirement, listing filters, home page filter, direct URL blocking)
- Job cancellation feature (grace period refunds for `in_progress` jobs)
- Kinglancer dashboard incomplete-profile banner
- `ConfirmModal` error prop

6. Platform Fees

- **Client pays:** budget + 2.5% platform fee
- **Kinglancer receives:** budget – 5% platform fee
- **Total platform take:** 7.5% of job budget
- Constants in `lib/stripe.ts`: `PLATFORM_FEE_RATE_CLIENT = 0.025` ,
`PLATFORM_FEE_RATE_KINGLANCER = 0.05`
- Auto-release: **5 working days** after work marked complete with no client response

7. Feature Status — Complete Inventory

✅ Authentication

- Email/password sign-up (multi-step: role → details → verify email)
- Google OAuth sign-in
- KingsChat SSO (cross-site POST → 303 redirect fix — see security notes)
- Forgot/reset password
- Email verification for non-Google users
- Onboarding page: role, bio (kinglancer required), services (at least one priced), phone, portfolio_url, cv_url

✅ Kinglancer Public Profile

- Route: `/kinglancers/[id]`
- **Completeness gate:** profiles are only visible if `bio.trim()` is non-empty AND at least one service has `rate > 0`. Incomplete profiles return 404 on direct URL access.
- Listing page (`/kinglancers`) also filters out incomplete profiles in JS.
- Home page "Top Kinglancers" component also filters out incomplete profiles.
- Kinglancer dashboard shows a red banner if profile is incomplete.

✅ Job Posting

- Clients can post public jobs or send direct requests to specific kinglancers.
- Categories, deadline, budget, rate type (fixed/hourly/daily).
- On post: bulk in-app notifications sent to top 50 matching kinglancers + email fan-out via `emailJobAlert`.
- For direct requests: only the invited kinglancer is notified.

✅ Job Application Flow

- Kinglancers apply to open jobs.
- Client receives in-app notification + email linking directly to `/dashboard/client/jobs/[jobId]`.
- Client can accept/reject applications.
- Accepted kinglancer receives "job awarded" notification.

✅ Direct Request Flow

- Client can invite a specific kinglancer.
- Kinglancer can accept/decline/counter-propose.
- On accept: status becomes `accepted_pending_payment`.
- Client pays → job goes `in_progress`.

✅ Payment Flow (Stripe Connect)

- Separate charges/transfers model.
- `payment_attempts` table tracks the PaymentIntent before finalization.
- On `payment_intent.succeeded` webhook: `finalizePaymentAttempt()` called.
 - Inserts a `transactions` row (status: `held`).
 - For applications: calls `selectApplicant()` to accept the winning application.
 - For direct requests: marks job `in_progress`.
- **Idempotency fix (staging only, needs to reach production):** if two concurrent webhook deliveries race, the second hits `transactions_job_id_unique` (23505). Now caught gracefully — returns 200 OK instead of 500. Stripe stops retrying.
- Express checkout (Apple Pay / Google Pay) — on `feat/express-checkout-wallets` branch, merged to `main`.

✅ Escrow / Release Flow

- Work submitted → client approves → `fireTransfer()` to kinglancer's Stripe Connect account.
- Auto-release cron: `/api/cron/auto-release` — runs via Railway cron service.
- `GRACE_PERIOD_MS = 2 * 60 * 60 * 1000` (2 hours) for in-progress job cancellation.

✅ Review System (Double-Blind)

- Both client and kinglancer submit reviews independently after job completion.
- Reviews are hidden until **both** submit, or after a 14-day timeout.
- Cron: `/api/cron/reveal-reviews` — reveals timed-out reviews.
- Pending reviews surface in the Action Centre with a countdown timer.

- Dashboard performance indexes in migration 025/026.

✅ Dispute System

- Client or kinglancer can raise a dispute on an `in_progress` or `completed` job.
- Admin receives notification.
- Admin resolves via `/admin/disputes`.

✅ Admin Panel

- Route: `/admin` (protected by passcode + HTTP-only cookie)
- Pages: `/admin/users`, `/admin/jobs`, `/admin/disputes`
- Pagination, search.

✅ Notification System

- In-app: `notifications` table, `NotificationBell` component polls for unread count.
- Email: via Brevo. Only fires when `ENABLE_EMAIL=true` AND `BREVO_API_KEY` AND `BREVO_SENDER_EMAIL` are set.
- Email helper functions in `lib/notifications.ts`:
 - `notify()` — creates in-app notification + optionally sends email
 - `emailJobAlert()` — email-only (no DB row), for fan-out on job post
 - `notifyNewApplication()` — client gets email with direct job link
 - `notifyJobAwarded()`, `notifyWorkSubmitted()`, `notifyPaymentReleased()`, `notifyDisputeRaised()`, `notifyPayoutReady()`, `notifyJobCancelled()`

✅ Job Cancellation

- Route: `POST /api/jobs/[id]/cancel`
- **open jobs:** bulk-rejects all applications + marks cancelled + notifies invited kinglancer (if direct request).
- **in_progress jobs:** checks transaction `created_at` within 2-hour grace period. If within: Stripe refund + update tx to `refunded` + cancel job + notify kinglancer. If outside: 409 `GRACE_PERIOD_EXPIRED`.
- UI: `CancelJobButton` client component on job workspace page.
- **Note:** This feature was built but is currently NOT on the staging or main branches (was reset). Needs to be re-applied.

✅ KingsChat SSO

- Cross-site POST from KingsChat hits `/auth/callback`.
- **Critical fix:** uses 303 redirect (not 307) so `SameSite=Lax` session cookies survive.
- **Security:** strictly requires `kcProfile.is_email_verified === true` before linking — prevents account takeover.
- Links KingsChat accounts to Supabase by email match.

✅ Stripe Connect Onboarding

- Kinglancers must complete Stripe Connect onboarding before receiving payouts.
- `stripe_onboarding_complete` boolean on profile.
- Webhook `account.updated` syncs payout status.
- After onboarding completes, automatically fires any pending unreleased transfers.

❌ Organisation Feature (PLANNED, NOT STARTED)

See Section 9 below.

8. Key File Locations

API Routes

Route	File	Description
POST /api/jobs	app/api/jobs/route.ts	Create job + fan-out notifications/emails
POST /api/jobs/[id]/cancel	app/api/jobs/[id]/cancel/route.ts	Cancel job with optional refund
POST /api/applications	app/api/applications/route.ts	Submit application
POST /api/profile/complete-onboarding	app/api/profile/complete-onboarding/route.ts	Save onboarding data
POST /api/webhooks/stripe	app/api/webhooks/stripe/route.ts	Stripe webhook handler
GET /api/health	app/api/health/route.ts	Health check
GET /api/cron/auto-release	app/api/cron/auto-release/route.ts	Auto-release escrow
GET /api/cron/reveal-reviews	app/api/cron/reveal-reviews/route.ts	Reveal timed-out reviews
GET /api/cron/cleanup-abandoned-checkouts	app/api/cron/cleanup-abandoned-checkouts/route.ts	Cancel stale PaymentIntents

Library Files

File	Description
lib/notifications.ts	All notification + email helpers
lib/db/payment-attempts.ts	PaymentAttempt DB operations + finalizePaymentAttempt()
lib/db/applications.ts	Application DB operations + selectApplicant()
lib/db/jobs.ts	Job DB operations
lib/db/transactions.ts	Transaction DB operations
lib/stripe.ts	Stripe helpers, fee constants, fireTransfer()
lib/stripe-connect.ts	Stripe Connect account creation
lib/admin-auth.ts	Admin passcode auth
lib/roles.ts	Role check helpers
lib/pagination.ts	Shared pagination utility
lib/supabase/service.ts	createServiceClient() — service role, bypasses RLS
lib/supabase/server.ts	createServerClient() — user session, respects RLS

lib/supabase/client.ts	Browser Supabase client
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Dashboard Pages

Route	File
/dashboard (shell)	app/(dashboard-shell)/layout.tsx
Client dashboard	app/(dashboard-shell)/dashboard/
Client job workspace	app/(dashboard-shell)/dashboard/client/jobs/[id]/page.tsx
Kinglancer dashboard	app/(dashboard-shell)/dashboard/kinglancer/page.tsx
Profile edit	app/(dashboard-shell)/dashboard/profile/page.tsx

9. Planned Features (Next Steps)

IMMEDIATE — Production fix (URGENT)

Problem: The webhook idempotency fix (commit `458019a`) is on `staging` but NOT on `main` (production). Production is currently in a Stripe webhook retry loop every time a payment succeeds.

Action:

```
git checkout main
git merge staging --no-edit
git push origin main
```

Note: Before merging, confirm the staging-only email features are ready to ship to production (they should be safe — they only activate when `ENABLE_EMAIL=true`).

HIGH PRIORITY — Test email notifications on staging

What to test:

1. Set `ENABLE_EMAIL=true` in Railway staging environment (and redeploy if not done already).
2. Post a new job as a client → check that kinglancers receive an email.
3. Apply to a job → check that the client receives an email with a direct link to the job page (`/dashboard/client/jobs/[jobId]`).

Files involved: `lib/notifications.ts`, `app/api/jobs/route.ts`, `app/api/applications/route.ts`

HIGH PRIORITY — Run migration 027 on staging and production

Run this SQL manually in Supabase SQL Editor for **both** staging and production databases:

```
create index if not exists idx_reviews_reviewer_job
on public.reviews (reviewer_id, job_id);

create index if not exists idx_reviews_reviewee_published_at
```



```
on public.reviews (reviewee_id, published_at desc)
where is_published;
```

File: supabase/migrations/027_review_query_indexes.sql

HIGH PRIORITY — Re-apply reset feature work to staging

The following features were built but then the staging branch was hard-reset (`git reset --hard 97c332b`) to match production. These need to be re-implemented:

1. Kinglancer Profile Completeness Gating

Rule: A kinglancer profile is "complete" if `bio` is non-empty AND at least one service in `services[]` has `rate > 0`.

Changes needed:

app/onboarding/page.tsx

- Add `bio` state (string, required for kinglancers, max 500 chars)
- Add "About you" textarea section before the services section (kinglancer-only)
- Validation: bio required + at least one service with `rate > 0`
- Pass `bio` in the API body

app/api/profile/complete-onboarding/route.ts

- Accept `bio` in request body
- Validate: bio not empty and `<= 500` chars for kinglancers
- Validate: at least one service has `rate > 0` for kinglancers
- Include `bio` in the DB update for kinglancer role

app/kinglancers/page.tsx

- `getKinglancers()` must select `bio`
- Filter results: `filter(k => k.bio?.trim() && rawServices.some(s => Number(s.rate) > 0))`

app/kinglancers/[id]/page.tsx

- After `notFound()` check, add completeness check:

```
if (!kinglancer.bio?.trim() || !rawServices.some(s => Number(s.rate) > 0))
  notFound();
```

components/home/TopKinglancers.tsx

- Fetch `bio` in select
- Filter out incomplete profiles with same rule
- Cache tag: `top-kinglancers`, 24h TTL

app/(dashboard-shell)/dashboard/kinglancer/page.tsx

- Compute `isProfileComplete = !!profile.bio?.trim() && services.some(s => Number(s.rate) > 0)`
- Show red banner when incomplete: `border-red-200 bg-red-50`

- Banner text: "Your profile is not visible to clients yet" + "Complete your profile →" link

2. Job Cancellation

New file: `app/api/jobs/[id]/cancel/route.ts`

```
POST /api/jobs/[id]/cancel
Auth: required, must be job owner (client_id)
```

Logic:

- If `status === 'open'` : bulk-reject applications + mark cancelled + notify invited kinglancer (direct request only)
- If `status === 'in_progress'` :
 - Get transaction → check `stripe_payment_intent_id` not null
 - Check `transaction.created_at` within `GRACE_PERIOD_MS = 2 * 60 * 60 * 1000` (2 hours)
 - If within grace: Stripe refund + update transaction to `refunded` + cancel job + `notifyJobCancelled()` to kinglancer
 - If outside grace: return 409 with `code: "GRACE_PERIOD_EXPIRED"`
- Other statuses: 400

New file: `app/(dashboard-shell)/dashboard/client/jobs/[id]/CancelJobButton.tsx`

- Client component
- Props: `jobId: string`, `status: "open" | "in_progress"`, `hasApplications?: boolean`
- Uses `ConfirmModal` with state-appropriate copy
- Redirects to `/dashboard/client/jobs` on success
- On grace period expired: shows inline error in modal

app/(dashboard-shell)/dashboard/client/jobs/[id]/page.tsx

- For `open` jobs: render `<CancelJobButton jobId={id} status="open" hasApplications={applications.length > 0} />` below applicants section
- For `in_progress` jobs: render `<CancelJobButton jobId={id} status="in_progress" />` in job workspace card

components/ConfirmModal.tsx

- Add optional `error?: string` prop
- Render red alert box inside modal above action buttons when `error` is set

DEFERRED — Organisation Feature

This is a planned new user type. Team message has been sent; waiting for responses.

Key agreed decisions so far:

- New role type named **"Organisation"** (not "Company", not "Business")
- Monthly subscription fee for access to the volunteer pool
- Verification against Companies House or Charity Commission register (admin manually certifies)
- Public organisation profile page
- **"Open to placements"** toggle on kinglancer profile (opt-in to volunteer work)
- Admin awards a certification badge to verified orgs (revocable)
- Disputes for volunteer placements go to admin

- 6-month maximum per placement
- Placement agreement auto-generated on both sides
- Experience portfolio on kinglancer profile (shows completed placements)

Not yet decided: specific subscription pricing, whether kinglancers can be paid by orgs at a reduced rate vs fully volunteer, exact UI for org discovery/search.

10. Known Bugs / Issues

Active Production Bug

- **Stripe webhook retry loop:** `payment_intent.succeeded` delivered twice concurrently → second delivery hits `transactions_job_id_unique` (23505) → webhook returns 500 → Stripe retries infinitely. **Fix is on staging (458019a), not yet on production.**
- **Impact:** Noisy logs, Stripe dashboard shows webhook failures. No double-charging occurs (constraint prevents it). The correct transaction record IS created.

Non-critical / Known

- Migration `027` not yet applied to any database — missing review query indexes. Queries still work, just slower at scale.
 - Profile completeness feature was reset from staging — incomplete kinglancer profiles are currently visible on the listing and home page.
 - Job cancellation feature was reset from staging — clients have no way to cancel jobs via UI.
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11. Stripe Payment Flow — Detailed

1. Client selects kinglancer (application or direct request)
2. POST `/api/payment-attempts/create`
 - `createPaymentAttempt()` – inserts `payment_attempts` row (`status: 'pending'`)
 - `Stripe.paymentIntents.create()` – creates PI
3. Client completes payment on `/checkout` page
4. Stripe fires `payment_intent.succeeded` webhook
5. POST `/api/webhooks/stripe`
 - `finalizePaymentAttempt(pi.id)`
 - a. Looks up `payment_attempts` by `stripe_payment_intent_id`
 - b. Checks for existing transaction on `job_id` (idempotency)
 - If exists + same PI: update to 'held', return `finalizedNow: false`
 - If exists + different PI: throw (different payment already funded job)
 - If 23505 on insert: catch gracefully, return `finalizedNow: false`
- (IDEMPOTENCY FIX)
 - c. Validates job state
 - d. For applications: calls `selectApplicant()` to accept winner
 - e. For direct requests: updates job to `in_progress`
 - f. Inserts transaction (`status: 'held'`)
 - g. Updates `payment_attempt` to 'succeeded'
 - h. Returns `{ attempt, finalizedNow: true }`
 - Notifies kinglancer they were hired
6. Client approves work (or 5-day auto-release timer expires)
7. POST `/api/jobs/[id]/approve` or cron triggers
 - Update transaction to 'released'

```
→ fireTransfer() – Stripe transfer to kinglancer's Connect account
  (amount = budget * (1 - 0.05))
8. Stripe fires account.updated webhook (after Connect onboarding)
→ If any pending released transactions, fire queued transfers
```

12. Cron Jobs

Configured in `railway.toml` as separate Railway services.

Cron	Schedule	Route	Description
auto-release	Every hour	/api/cron/auto-release	Releases escrow after 5 working days
reveal-reviews	Daily	/api/cron/reveal-reviews	Reveals reviews after 14-day timeout
cleanup-abandoned-checkouts	Every 30 min	/api/cron/cleanup-abandoned-checkouts	Cancels stale PaymentIntents

All cron endpoints require `Authorization: Bearer ${CRON_SECRET}` header.

13. Security Notes

KingsChat SSO

- Callback uses **303 redirect** (not 307). 307 preserves the POST method, which blocks SameSite=Lax cookies. This fix is in `lib/kingschat-auth.ts`. Any future cross-site POST handlers must also use 303.
- **Email verification required:** `kcProfile.is_email_verified` must be `true` before linking accounts. Prevents account takeover where someone with an unverified KingsChat account could hijack a KingsHire account.

RLS

- All Supabase tables have Row Level Security enabled.
- Service role client (`createServiceClient()`) bypasses RLS — only used in server-side API routes, never exposed to client.
- Browser client uses `createBrowserClient()` which respects RLS.

Admin

- `/admin` routes protected by `ADMIN_PASSCODE` + HTTP-only cookie (signed with `ADMIN_SESSION_SECRET`).
- Admin routes in `app/admin/(protected)/` — layout checks cookie on every request.

14. How to Run Locally

```
npm install
npm run dev    # starts Next.js dev server on :3000
```

Required: `.env.local` with all env vars from Section 4.

For Stripe webhooks locally: `stripe listen --forward-to localhost:3000/api/webhooks/stripe`

15. Recommended Next Agent Prompt Prefix

When starting a new chat, paste this context:

This is the KingsHire (kingshire-v3) project. Next.js 16.2.4 App Router, Supabase, Stripe Connect, Brevo email, deployed on Railway. The AGENTS.md says "This is NOT the Next.js you know — read node_modules/next/dist/docs/ before writing code." Current branches: `staging` is ahead of `main` by 4 commits (webhook idempotency fix + email fixes). The most urgent task is merging staging to main to fix a production webhook retry loop. See `/docs/HANDOVER.md` in the repo for the full project state.